MCT-454L – Mobile Robotics

Lab-05

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Objective

The lab session was conducted about the usage of Gazebo and how to instegrate it with RViz using the TurtleBot3 platform to simulate the robot in simulated environment.

Procedure

1. First, install the Turtlebot3 and Gazebo packages via command "sudo apt install ros- humble-turtlebot3* ros-humble-gazebo-ros-pkgs".
2. Open terminal and set the Turtlebot3 model by exporting wafle or burger.
3. Then, source ROS2 workspace and launch the Gazebo simulator using the command "ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py".
4. Again open new terminal and generate the RViz software using the following command:

   "ros2 launch turtlebot3_cartographer cartographer.launch.py use_sim_time:=true".
5. Now exploring some features of RViz such as Add button to select plugins i.e, LaserScan, Path, Map, TF, Odemetry etc.
6. Adjust the frame settings under "Global Options" to odom or map and set the Fixed Frame to map to align all data with the world frame.
7. Open a new terminal and source the ROS 2 workspace if needed.
   - Run the keyboard teleoperation node: ros2 run turtlebot3_teleop teleop_keyboard
   - Use the following keys to control the robot:

W → Forward	**D** →Turn right
S → Backward	**Q** → Increase speed
A → Turn left	**Z** → Decrease speed
S → Stop Forcefully	

   - Use the following command to record all topics: ros2 bag record -a
   - When the robot cover the all whole bounded area and localize it in map, then save the map via: "cd mkdir maps ros2 run nav2_map_server map_saver_cli -f maps/my_map".